

Atari 2600 in Minecraft

Gamer, Seth Bling builds the ancient Atari gaming in Minecraft console using thousands of command blocks. <http://dgit.in/AtriMncrft>



Microsoft Build 2017

Microsoft announces that its 2017 annual build developer conference will take place 10th-12th May, 2017 <http://dgit.in/MSb2017>

Feature

every movie or product in recommended section piques your interest. That's a very benign implementation of machine learning.

Milestones in 2016

They can beat you are your own game

2016 began with a flurry of news articles lauding the numerous victories of AlphaGo, a Google Deepmind computer program which can play the board game Go. It started by taking baby-steps, learning the game and its different strategies and then moved on to besting humans in October 2015 without handicaps. Five months down the line in March 2016, it had won in a best-of-five against Lee Sedol. For those who're coming across this name for the first time, Lee Sedol is a 9-dan (top Go world rank) professional player. In just five months, using current gen technology, an AI had become the very best in the human scale.



THE match: AlphaGo vs Lee Sedol

They can speak your language

Another key development was that of Google Neural Machine Translation, which is now used for Google's Translate service. With a little help from the Google Brain team, the folks behind the project were able to reduce translation errors in complex languages like Mandarin by 60 per cent compared to their previous phrase-based algorithms. This was in September 2016, and in just one month, the researchers behind the project have announced work on Zero-Shot Translation. Till now, all the 103 languages that Google Translate had to work on had separate computing machines. With Zero-Shot Translation, they aim to do away with all of them and replace that with just one

extremely powerful system. They've already put the system in practice for a few languages and are scaling it slowly.

They can single you out

We've spoken about games and languages, both key domains for ML. Another such key domain is that of visual recognition. Industrial applications, like that of security, which have hundreds of cameras tied into one computer, which analyses each video feed frame-by-frame, have already been around for a while. Human behavioural traits are now being learnt by these systems to mark individuals out from the crowd. There's even a startup called AllGoVision which has already put into production a system that can detect loitering, tailgating, suspicious objects and more.

To take things to the extreme end, scientists have developed a neural network that can identify criminals just by looking at photos. Students at the Shanghai Jiao Tong University fed 1,586 photographs of men, half of which were criminals, to a system which then learnt to identify criminals with a 89.5 per cent accuracy. No background data was provided whatsoever, so the only metric were their faces.

They can read your lips

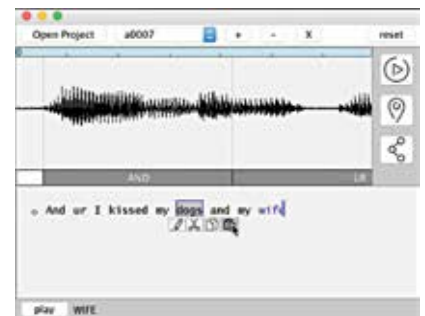
The folks at the University of Oxford seem to have grown tired of the BBC anchors' voices, so much that they came up with the 'Watch, Listen, Attend and Spell' neural model. Be it just plain audio, or video or both, this machine learning model can accurately transcribe speech into characters with a higher accuracy than professional lip-readers. The system was so good, that it surpassed all previous benchmarks for lip-reading based on the same data sets.



Lip reading accuracy has improved

They can speak for you

Google's been keeping Brain quite busy – they recently published a paper introducing WaveNet, a deep neural network for generating raw audio forms. This is simply a text to speech system, except that it doesn't sound robotic anymore. Parametric systems, i.e. the ones you're all quite familiar with, be it at public airports, or your operating system's friendly voice, uses snippets of audios which are stitched together to form words. These sound very monotonous but when you get a neural



Adobe's VoCo is PhotoShop for audio

network to analyse millions of audio files, it can generate the same sentence with a much higher resolution. Imagine current gen systems as having individual audio files for each letter. Based on their position in a word or sentence, a particular audio file would be played. Now this is the equivalent of a highly pixelated image. Now imagine if more audio files were used per alphabet, you'd have tonality. Now go one level deeper with even more files per sound. You now have WaveNet, which speaks very much like a human being.

One thing that politicians will be blaming each time they're "misquoted" will be Adobe's VoCo. Why? Well, it's photoshop ... for audio. Just like WaveNet, it can take an audio stream from a person, transcribe it, allow you to edit the sentence and play it back ... in their own voice! This technology was expected but no one expected it to materialise this quick. While it does raise several concerns about ethics, and how digital media could no longer be used as evidence in the courtroom, it sure does make for a really cool tech-